

PLANETARY PS

AN INDEPENDENT PLANETARY PS READOUT

The Colorado River after 2026

DATE **October 26, 2026**

COVERS **Where the shortfall lands**

SUMMARY

The Colorado River Basin is in crisis, and the new federal rules cover only the next two years

THE RIVER
Forty million people depend on it

Agreements promise 16.5 million acre-feet a year; the river averaged 12.4 million from 2000 to 2024.

- Las Vegas gets about 90% of its water from it, and Phoenix about 40%.
- Each degree Celsius of warming cuts its flow by about 9%.

HOW THE BASIN HAS COPED
Three ways, each with a cost

Since 2000 the basin has covered the gap three ways, and each now shows a cost.

- Drawing down reserves: groundwater is two-thirds of the water the basin has lost since 2002.
- Shifting the shortage: Central Arizona Project use is down 40%, and the Salton Sea is shrinking.
- Building new supply: Phoenix's first purification plant equals about 1% of Arizona's cut.

THIS YEAR
Record lows and imposed rules

Both reservoirs hit record lows, together holding under a quarter of their 1983 peak.

- After the states failed to agree, the federal government imposed cuts for 2027 and 2028; Nevada sued.
- The Lower Basin's apportionment falls to 6.25 million acre-feet, above what it used in 2025.

WHAT COMES NEXT
The long-run split

Who uses less for good goes to the next round of rules, which must start by October 2027.

- Even cuts near 2.4 million acre-feet a year would leave Lake Mead below a critical level more than a fifth of the time.
- None of the six proposals on the table adds water to the river.

Sources: Record of Decision and Operating Guidelines (August 2026); CRS; Reclamation data; detail on each slide.

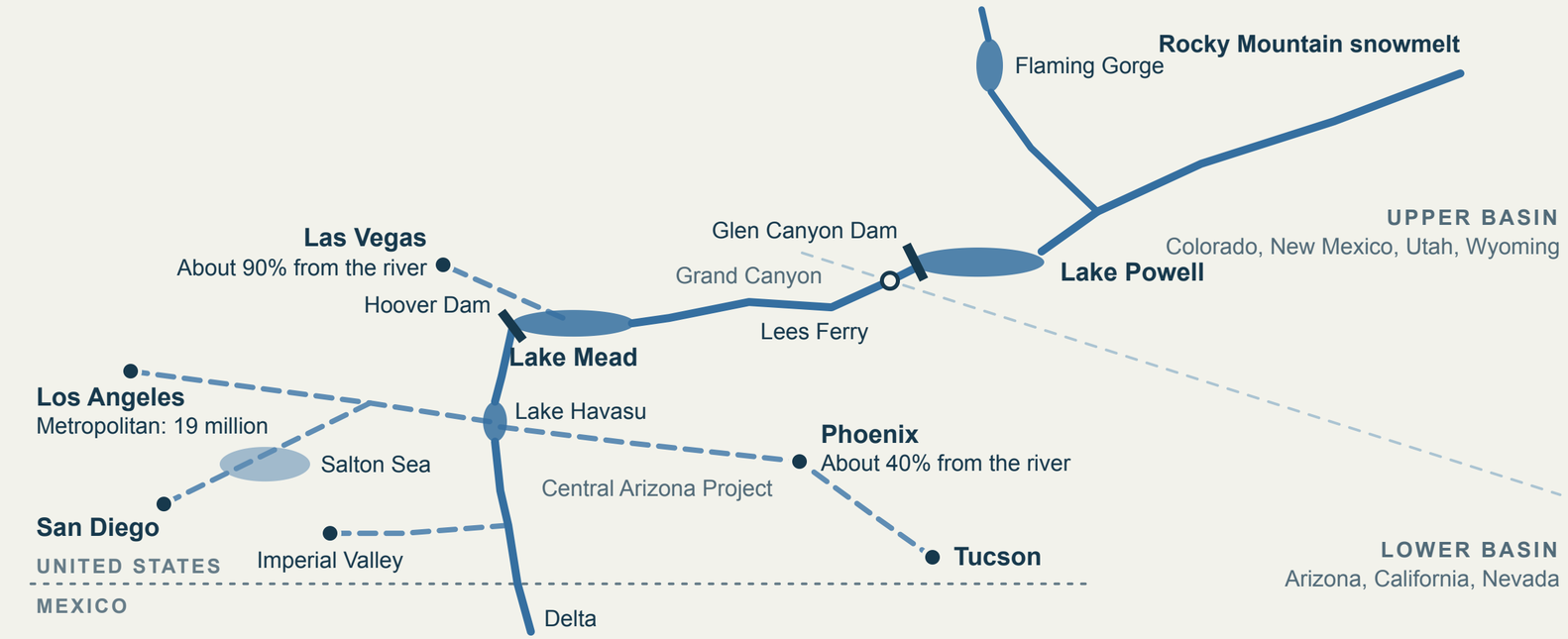
One river supplies 40 million people, 5.5 million acres of farms, 30 tribes, and Mexico

It starts as Rocky Mountain snow; two big reservoirs store it, their dams make power, and canals carry it to cities and farms.

The Colorado River and where its water goes

Main reservoirs, dams, and canals; Metropolitan is a Southern California water wholesaler

SCHEMATIC, NOT TO SCALE



WHO DEPENDS ON IT

40M people in seven states

5.5M acres of farmland

30 tribal nations, and Mexico under a 1944 treaty

WHAT ELSE RIDES ON IT

California grows most of the nation’s winter vegetables. Bloomberg Intelligence puts the exposed regional economy at $1 trillion.

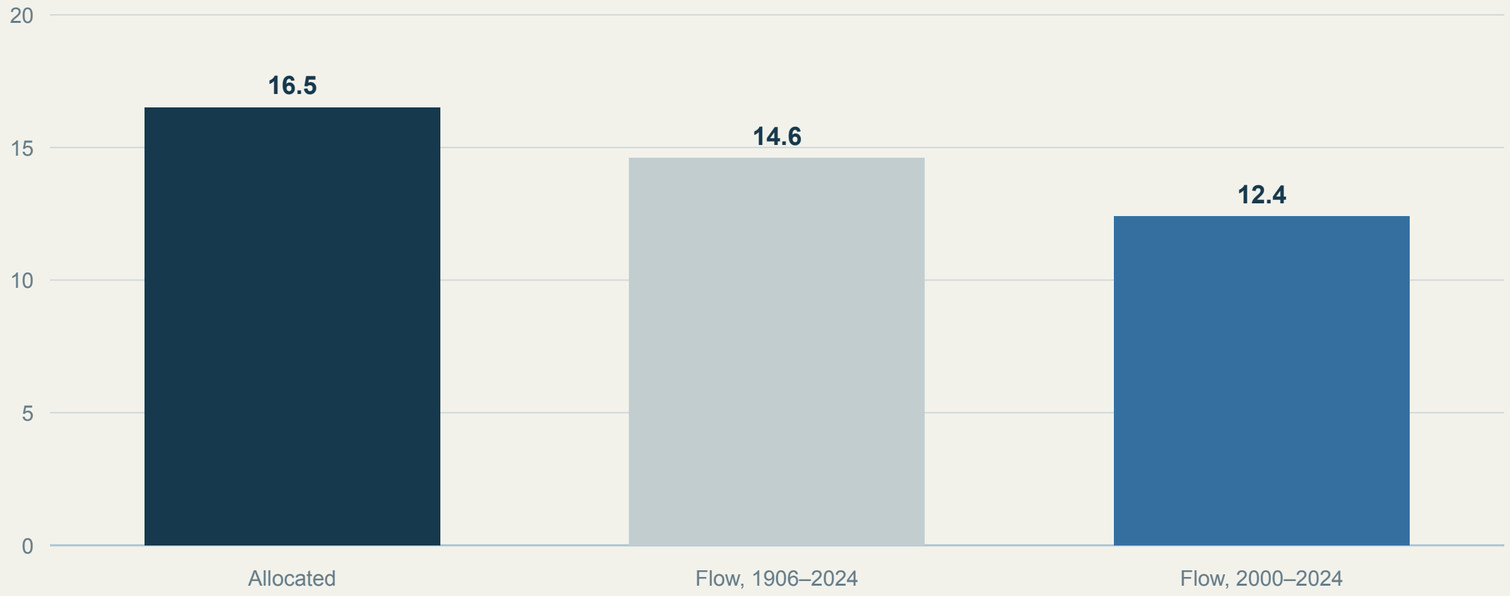
Sources: Record of Decision (August 2026), section 3.1; the Guardian (October 1, 2026); Bloomberg (October 19, 2026); Bloomberg (July 31, 2026), citing Bloomberg Intelligence; CalMatters (August 2026). Map drawn by Planetary PS.

Agreements from 1922 and 1944 promise 16.5 million acre-feet a year from a river that averaged 12.4 million from 2000 to 2024

The compact split the river between the four upstream and three downstream states; a 1944 treaty added Mexico. Flows from 2000 to 2024 ran about 15% below the long-run average.

Allocations against natural flow

Million acre-feet a year; natural flow at Lees Ferry, averages for the periods shown



WHAT THE ALLOCATION ADDS UP TO

Two basins and Mexico

- Upper Basin 7.5, Lower Basin 7.5, and Mexico 1.5 million acre-feet a year
- Reservoir evaporation, 1.4 million acre-feet in 2024, comes on top

READ WITH CARE

Allocations are legal entitlements; natural flow estimates what the river would carry without dams. The dry 2025 and 2026 would lower the averages.

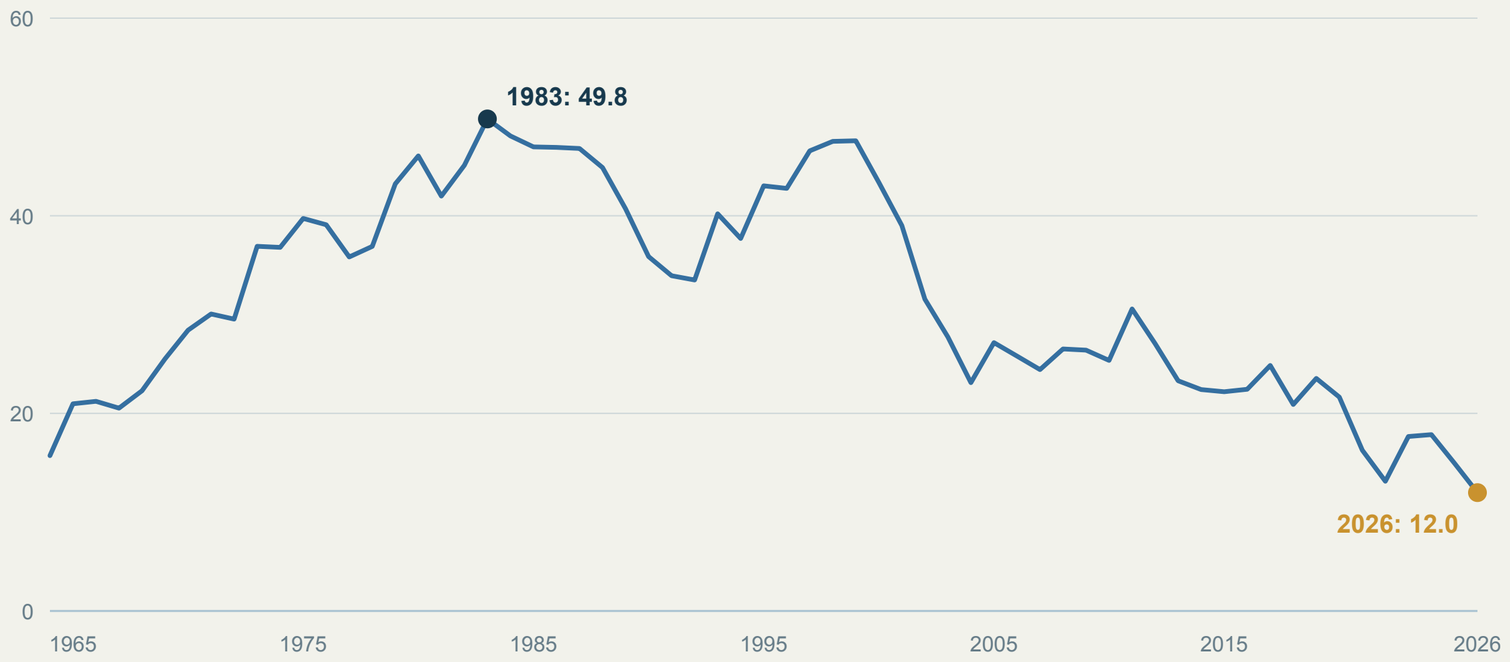
Sources: CRS R45546 (October 1, 2026) for the compact, the treaty, and natural flow; evaporation from Schmidt et al., Part 2 (September 2, 2026), Table 1, citing Reclamation. Calculations: the 16.5 total and the 15% gap.

Both reservoirs hit record lows this year, together holding under a quarter of their 1983 peak

Lake Powell and Lake Mead held 12.0 million acre-feet at the end of September, against 49.8 million in 1983.

Lake Powell plus Lake Mead, storage at the end of each water year

Million acre-feet, September 30; Powell began filling in 1963



RECORD LOWS

Both lakes are below their earlier lows

- Powell: 3,516.6 feet on September 15, 27 feet above the level where its turbines stop
- Mead: 1,037.2 feet on October 22, below its 2022 low

CONTEXT

Combined storage was the lowest since 1957.

Sources: Reclamation reservoir data, Lake Powell (site 919) and Lake Mead (site 921), through October 22, 2026; Operating Guidelines, section 5.1. Calculations: combined totals and comparisons.

The federal government imposed two years of cuts in August after the seven states failed to agree

The cuts fall on the three downstream states. The four upstream states face none they must make.

THE CUT

1.25 million acre-feet a year in 2027 and 2028

Arizona 760,000, California 440,000, and Nevada 50,000. The framework allows up to 3 million a year through 2036.

UPSTREAM

Contributions are voluntary

Under the longer framework, Colorado, New Mexico, Utah, and Wyoming may contribute up to 200,000 acre-feet a year. The two-year rules set no amount.

THE CHALLENGE

Nevada sued, and Arizona moved to join

In the worst case, Nevada’s share could be cut by as much as 70%. Arizona’s governor says the Upper Basin owes Arizona deliveries.

HOW THE DECISION WAS MADE

Jun 2023	Interior starts the process to replace the 2007 rules.
Nov 2025	The states miss the first deadline for a deal.
Jan 2026	Draft environmental review, with no preferred plan.
Feb 2026	The states miss the second deadline.
Jul 31	Final environmental review.
Aug 21	Interior signs the decision and the two-year rules.
Aug 24	Nevada sues to set the decision aside.
Sep 2	The US and Mexico agree on Mexico’s cuts through 2028.
Oct 1	**The new rules take effect.**
Oct 16	Arizona moves to join Nevada’s suit.

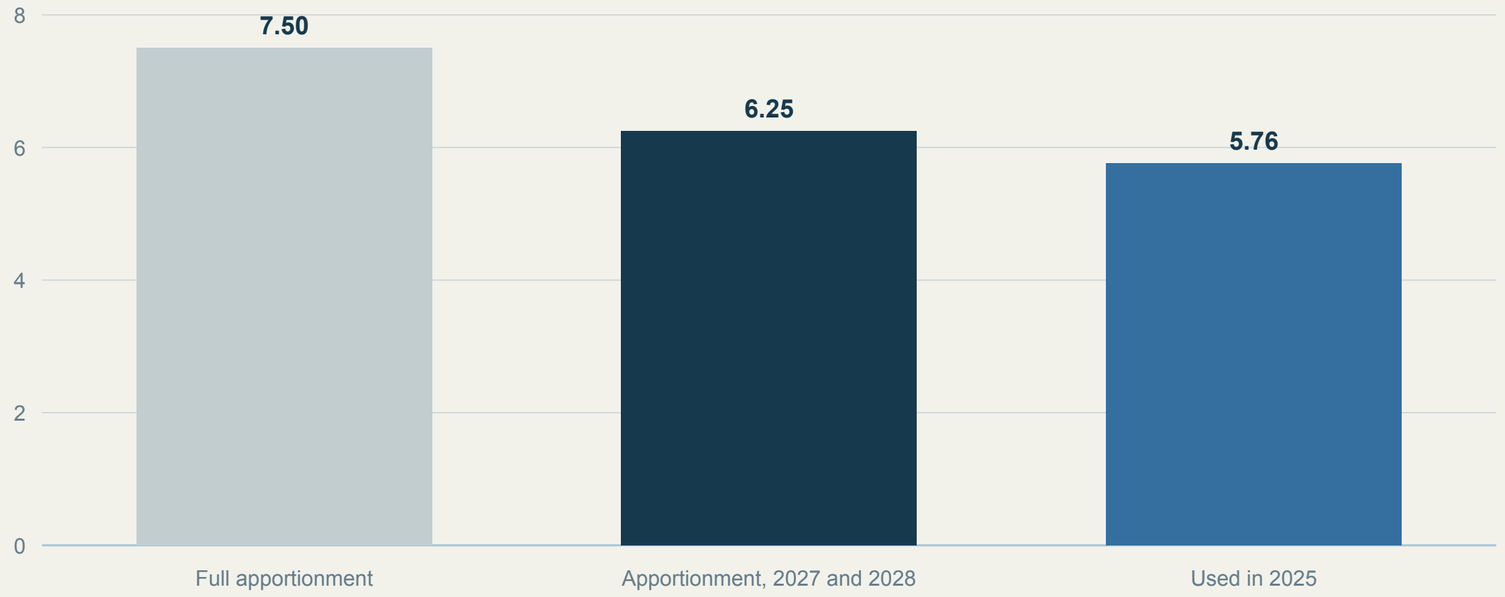
Sources: Record of Decision, Table 1 and section 5; Operating Guidelines, section 5.3; Western Water (August 21, 2026); Summit Daily on the missed deadlines; KUNC (August 25, 2026) and Bloomberg (October 19, 2026) on Nevada; WaterWorld on Minute 334; azfamily (October 16, 2026) and the Guardian (October 17, 2026) on Arizona.

The new rules cut the Lower Basin's apportionment to 6.25 million acre-feet, above the 5.76 million it used in 2025

Use fell in 2025 mostly through conservation agreements that leave water in Lake Mead. Stored savings can offset part of the cuts, and the rules seek 0.7 million acre-feet of voluntary conservation over 2026 to 2028.

Lower Basin use against its entitlement

Million acre-feet a year; Arizona, California, and Nevada combined



WHAT MADE 2025 LOW

Mostly water left in Lake Mead

- 1.13 million acre-feet of system conservation
- 0.53 million in cuts required by earlier rules
- 0.09 million of other conserved water

READ WITH CARE

A June dry-year scenario puts basin-wide use 2.59 million acre-feet above supply, before the August rules.

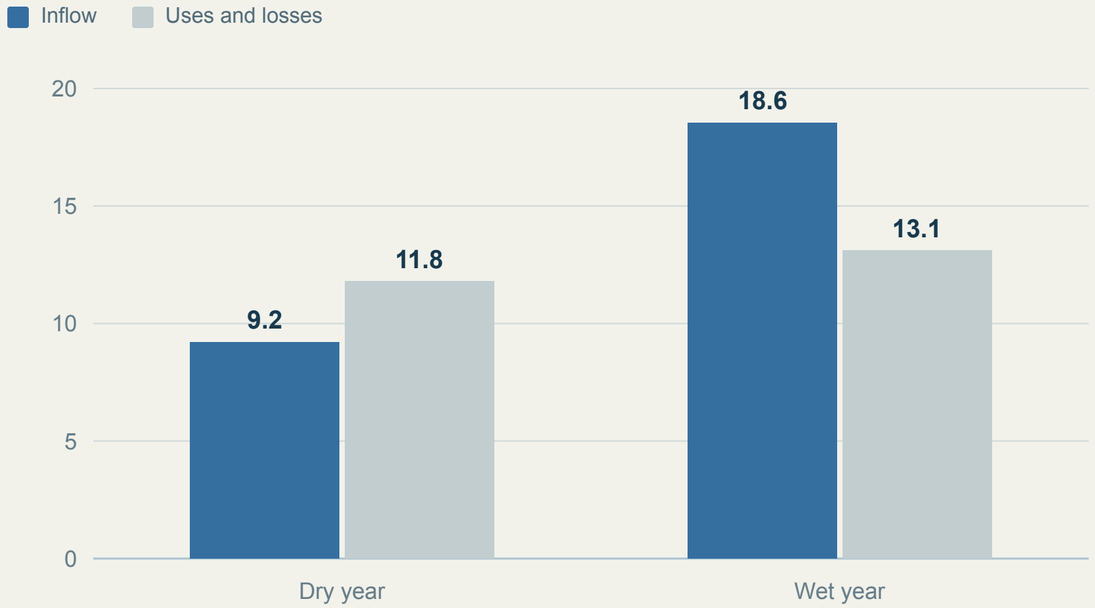
Sources: Operating Guidelines, sections 5.3.A and 5.3.B; Reclamation, Colorado River Accounting and Water Use Report, 2025 (May 2026), Table 11; Castle et al., June 1, 2026 update, appendix. Calculations: the splits are sums of Table 11.

Warming has cut the river's flow and widened its swings: next year could bring 9 or 19 million acre-feet

Each degree Celsius of warming cuts flow by about 9%. In researchers' June scenarios, a dry year leaves 3.6 million acre-feet of usable storage, and a wet one would provide "only about two years of cushion." These are scenarios, not forecasts.

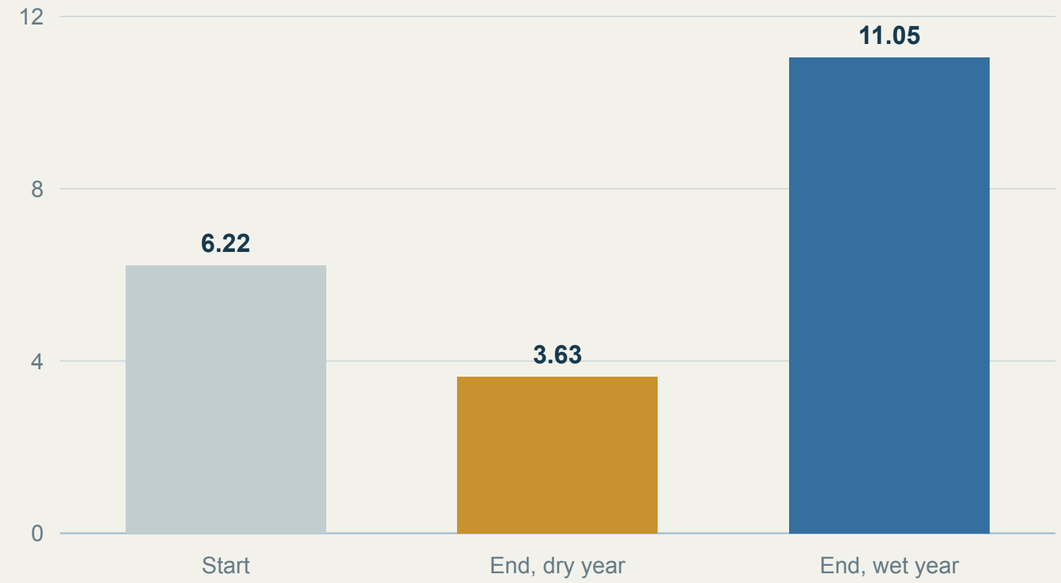
Water year 2027 scenarios: water in and water out

Million acre-feet; dry case like 2025, wet case like 2023



Usable storage, start and end of water year 2027

Million acre-feet above protected levels in Powell, Mead, and Flaming Gorge



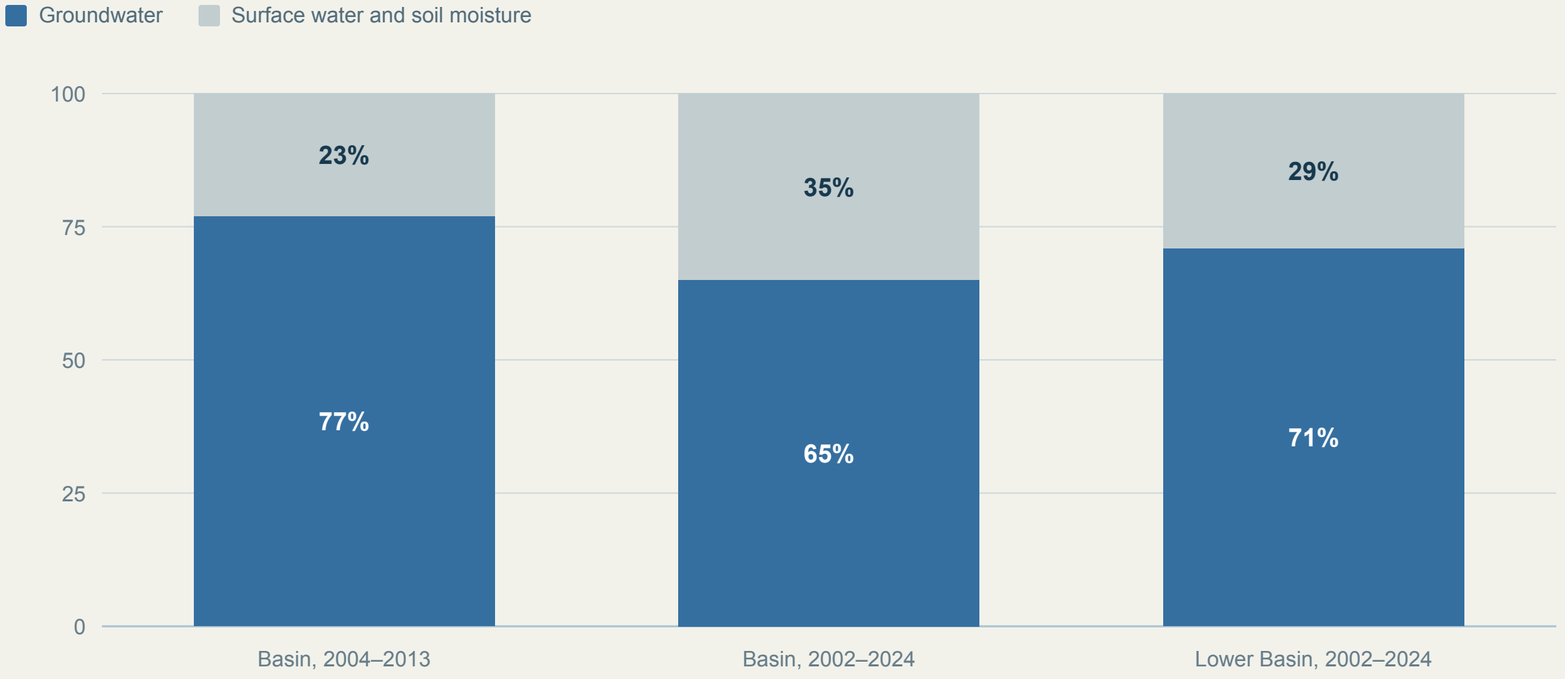
Sources: Castle et al., Colorado River Basin Storage Continues Slide Toward System Crash (June 1, 2026), scenarios and appendix, built on May forecasts before the August rules; Milly and Dunne, Science (2020), 9.3% per °C; Udall and Overpeck (2017). Usable storage is the authors' measure above 3,500 feet at Powell, 975 feet at Mead, and 0.5 million acre-feet at Flaming Gorge.

When the river comes up short, users pump: groundwater is two-thirds of the basin's water loss since 2002

Satellites measure all the water the basin has lost. Most of it came from aquifers, the basin's other reserve.

Groundwater share of the basin's lost water storage

Percent of total terrestrial water storage lost; GRACE satellite estimates



WHY IT MATTERS

Pumping hides the shortfall

- Of 52 cubic kilometers lost from 2002 to 2024, 34 were groundwater
- Farms that lost river water have turned to groundwater, which does not replenish, says ASU's Sarah Porter

READ WITH CARE

Groundwater is total loss minus surface water and soil moisture, so uncertainty is wide.

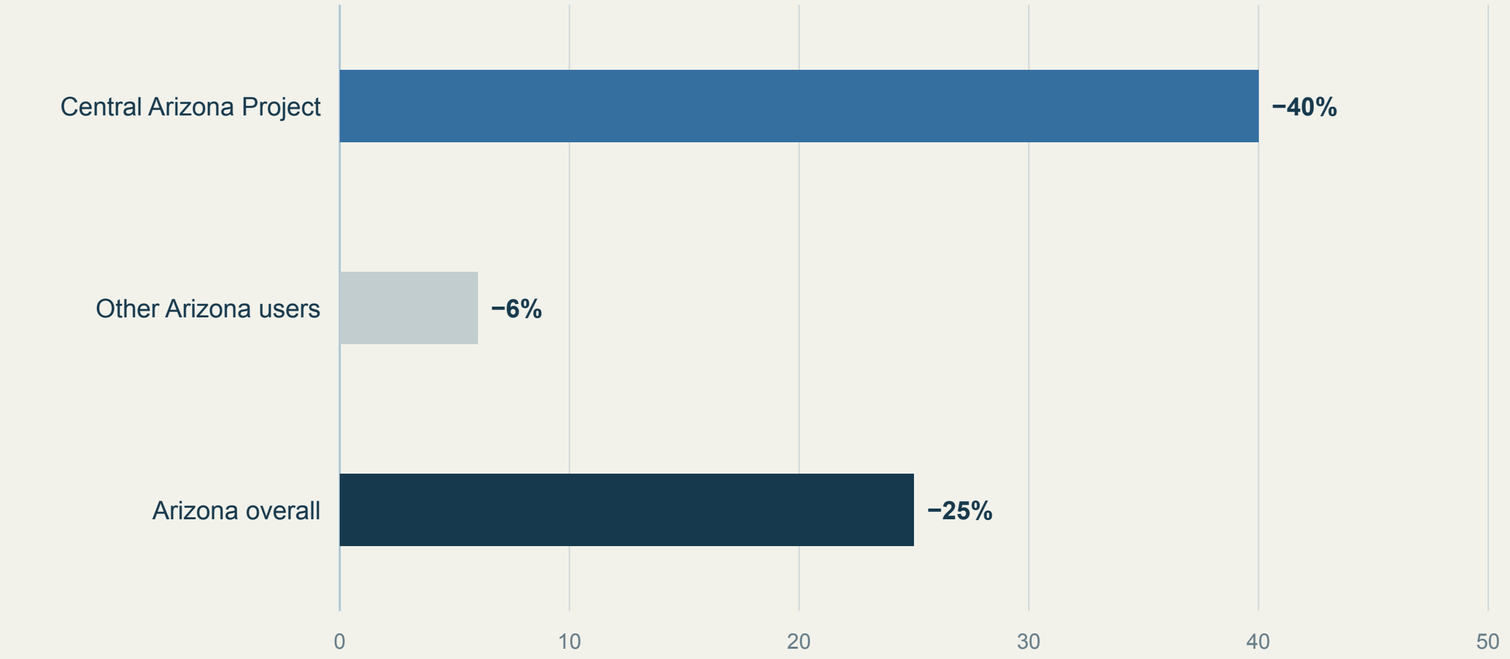
Sources: Castle et al., Geophysical Research Letters (2014); Abdelmohsen et al., Geophysical Research Letters (2025), April 2002 to October 2024; Bloomberg (July 31, 2026) on Porter.

Since 2023 Arizona’s cuts have fallen mostly on the Central Arizona Project, junior to California since 1968

The canal carries river water hundreds of miles to Phoenix, Tucson, and tribes.

Change in Arizona’s use of river water

Percent, 2023–2026 against 2010–2022



WHO IS JUNIOR

The canal is cut before California’s farms

- Its 1968 authorization made it junior to California’s users
- Arizona has disputed when that priority applies

NEXT YEAR

The City of Phoenix expects about 50% less water to flow through the canal.

Sources: Schmidt et al., Part 2 (September 2, 2026); CRS R45546 on the junior priority; E&E News (December 2022) on Arizona’s reading of it; the Guardian (October 1, 2026), quoting the City of Phoenix.

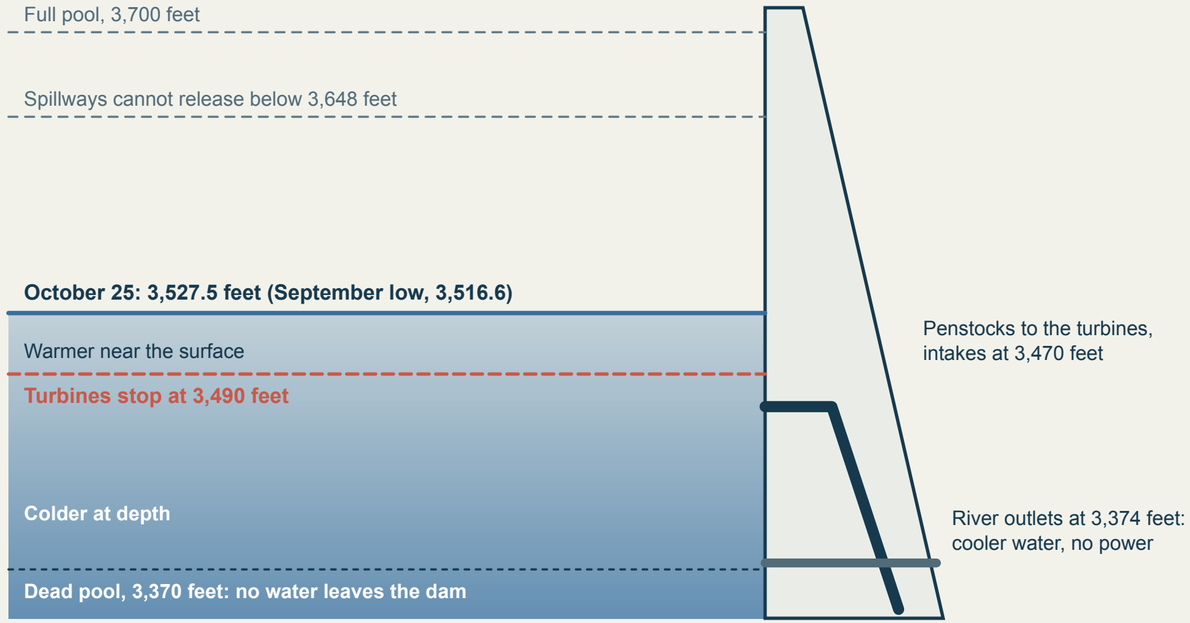
At Lake Powell's current level, the cold releases that keep invasive bass out of the Grand Canyon cost hydropower

Cooler water sits deep, near river outlets about 100 feet below the penstocks, and releasing it there bypasses the turbines.

Glen Canyon Dam and Lake Powell

Elevation in feet; the dam continues below 3,340 feet, not shown

NOT TO HORIZONTAL SCALE



THE TRADE-OFF

Cold water costs power

- Glen Canyon makes about 75% of the power in its federal system
- Cold releases in 2025 cost $6.5 million in replacement power

IN 2026

Reclamation forwent them, citing "significant hydropower generation concerns." Lees Ferry reached about 70°F, above the 60°F at which smallmouth bass spawn.

Sources: Reclamation technical memo (March 26, 2024); Reclamation (August 5, 2025); Reclamation reservoir data (site 919, provisional daily data); WAPA comment on the Draft EIS; WAPA (December 11, 2025); Circle of Blue (September 16, 2026). Diagram drawn by Planetary PS, to vertical scale.

Children living near the shrinking Salton Sea show slower lung growth, a 2026 study found

The study of 369 children found an association. It does not tie the effect to any one water deal.

1

Farm water moves to cities

Since 2003, Imperial Valley water has been transferred to San Diego and other cities.

2

Farms cut more

California farm districts are cutting more than 700,000 acre-feet over four years under paid conservation.

3

Inflows to the sea fall

Pacific Institute expects significant reductions in what reaches the Salton Sea.

4

The lakebed is exposed

The federal government owns more than 100,000 acres of it, and wind lifts its dust.

5

Children’s lungs grow less

Children living within 11 km of the sea had slower lung-function growth.

THE TRADE-OFF

The river’s ledger improves; the exposed lakebed stays put

Water saved in the Imperial Valley helps hold up Lake Mead. The dust stays in Imperial County.

Sources: Pacific Institute comment on the Draft EIS (March 2, 2026), on farm cuts, inflows, and federal lakebed ownership; JAMA Network Open (April 3, 2026), USC, UC Irvine, Comité Cívico del Valle, and the University of Washington. The transfer since 2003 is the Quantification Settlement Agreement.

Phoenix-area growth now needs water beyond groundwater, and its first purification plant equals about 1% of Arizona's cut

3.6 million

Acre-feet short

Phoenix-area groundwater demand left unmet over 100 years in the state's 2024 update, down from 4.9 million in 2023.

80,000

Homes already certified

Since 2023 new subdivisions cannot rely on groundwater alone.

825

Homes in the first farm-to-city approval

Under the 2025 Ag-to-Urban law, homes can replace retired farmland.

1%

Phoenix's first advanced purification plant, against Arizona's 760,000 acre-foot cut

Nearly 7 million gallons a day when it opens next year, about 7,800 acre-feet a year. Phoenix plans three and is spending billions to depend less on the river.

Sources: Arizona Department of Water Resources, as reported by AP (June 1, 2023) and AMWUA (December 3, 2024); Office of the Governor (December 9, 2025); the Guardian (October 1, 2026) and KTAR on the plant. Calculations: acre-feet a year and the 1% share.

Who pays for the shortfall is now a question for voters, bond investors, courts, and power customers

VOTERS

Water is the top issue for the first time

A June survey by the Center for the Future of Arizona, released in August, put long-term water supply first.

No. 1

voter concern in Arizona, 2026

BOND INVESTORS

S&P cited regulatory uncertainty

It rated Las Vegas's first water bonds since the cuts AA+, citing the district's record of curbing use as it grew. The district is also suing.

$580M

bond sale, October 2026

THE COURTS

Nevada cites its economy

Its suit argues Interior did not weigh the cost of deeper cuts to Southern Nevada's economy. Arizona has moved to join.

$180B

Southern Nevada economy, as cited in the suit

POWER CUSTOMERS

Protecting fish has a power bill

When cold-water releases ran in 2025, the federal power marketer paid to replace the lost hydropower.

$6.5M

replacement power, 2025

ALREADY SPENT

Las Vegas spent $1.3 billion on an intake nearly 200 feet deeper in Lake Mead and a new pumping station.

Sources: PBS NewsHour and Center for the Future of Arizona (August 14, 2026); Bloomberg (October 19, 2026) and The Bond Buyer on the Las Vegas Valley Water District sale; KUNC (August 25, 2026) on Nevada's suit; WAPA (December 11, 2025) on replacement power; the Guardian (October 1, 2026) on the intake.

The proposals split on who cuts use, whether savings last, and who controls the water saved

None of the six adds water to the river. The last column shows whether the August rules already provide for each.

	PATHWAY, AND WHAT IT CHANGES	WHO PROPOSES IT, AND THE NUMBERS	IN THE RULES?
1	**Cut use in step with the river** HOW MUCH IS USED	Seven NGOs, the Sonoran Institute, and researchers, one proposing a hard cap revisited every five years: about 3 million acre-feet in critical years.	PARTLY
2	**Savings pools in both reservoirs** WHEN SAVED WATER IS USED	Most groups, with access for tribes and Mexico. The ten-year framework allows up to 3.0 million acre-feet in Powell and 8.0 million in Mead.	ALLOWED
3	**Lasting money for conservation** WHO PAYS FOR CUTS	70 groups asked Congress for at least $2 billion. Farm conservation has averaged about $417 an acre-foot.	PENDING
4	**Water for the environment** WATER LEFT FOR NATURE	Conservation groups model Delta flows of 45,000 acre-feet a year, up to 135,000 in one year, and seek cooler Grand Canyon releases.	PARTLY
5	**Rethink Glen Canyon Dam** WHERE WATER IS STORED	Reach the 5.6 million acre-feet stored below the turbine intakes, or bypass the dam; others want Powell kept high.	NOT IN RULES
6	**Tribal Nations in governance** WHO DECIDES	Seven NGOs ask for structured, meaningful Tribal participation; the rules commit to consultation.	CONSULTED

WHERE THEY SPLIT

Who cuts: Nevada wants the Upper Basin to share; groups differ on paying or requiring farms to use less, and cattle-feed crops take about 62% of farm water. Whether paid savings last. Who controls saved water. A lower Powell or a cold river.

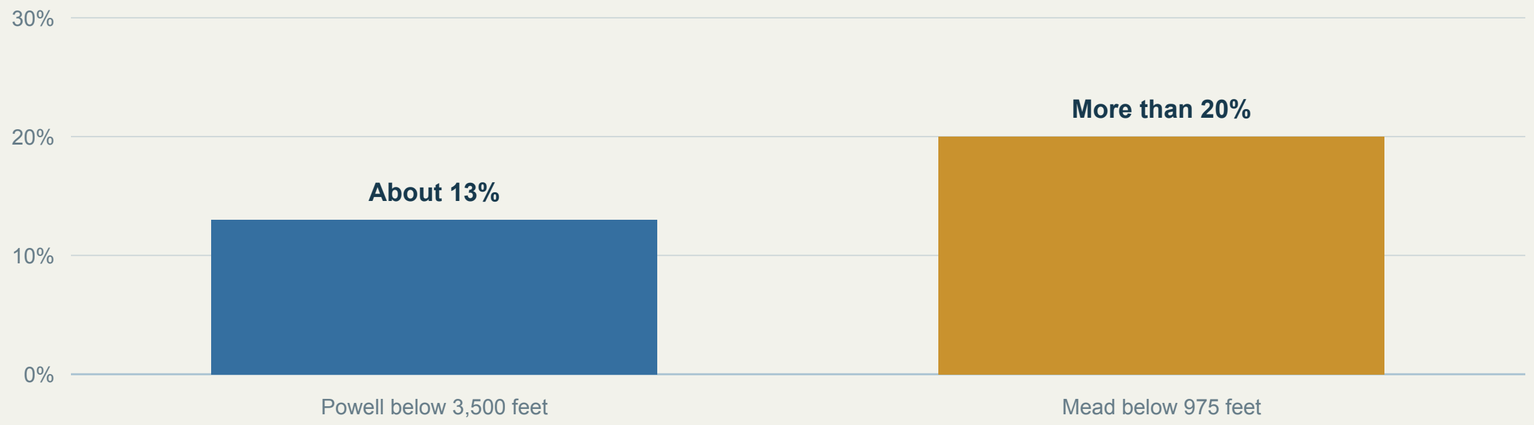
Sources: joint comment of seven NGOs (March 2, 2026); Cooperative Conservation alternative (2024); Sonoran Institute; 70-group letter, as reported (May 2026); UC Riverside (2025) and KSL (October 13, 2025); Pacific Institute; Glen Canyon Institute; Grand Canyon Trust; High Country News; NAU; KUNC; status from the Record of Decision. Status is a draft reading for discussion.

Even cuts near 2.4 million acre-feet a year would leave Lake Mead below a critical level more than a fifth of the time

Pacific Institute’s reading of the federal modeling, even assuming much better starting levels. The federal plan itself noted that cuts nearly twice what the states proposed could be needed.

Share of the next 20 years below critical elevations

Percent of time; most flexible draft alternative, about 2.43 million acre-feet of average cuts



REPLACING THE RIVER

Farm conservation programs averaged about $417 an acre-foot. New supply projects averaged more than $2,400.

QUESTIONS THE PROPOSALS LEAVE OPEN

Is it enough?

- What holds use down once federal money ends?
- Where does saved water go, without a defined right?
- Do surface cuts move the draw underground?
- Which comes first, a lower Powell or a cold river?
- What is the river worth, against the cost of replacing it?

Sources: Pacific Institute comment on the Draft EIS (March 2, 2026), citing the Draft EIS; Bloomberg (July 31, 2026); UC Riverside (2025) and KSL (October 13, 2025), on 462 Reclamation-funded projects, 2004 to 2024; Abdelmohsen et al. (2025); Grand Canyon Trust and Glen Canyon Institute. Replacement framing adapted from Eric Wilburn (October 15, 2026). The questions are a draft reading for discussion.

This winter’s snow, the lawsuit, Congress, and the next rules will decide the next two years

THE SNOWPACK

April forecasts set 2027 releases

In June scenarios, a dry year leaves 3.6 million acre-feet of usable storage. A wet one gives about two years of cushion.

3.6

MAF usable after a dry year

THE COURTS

Nevada and Arizona against Interior

Interior says litigation won’t add water. California says a lasting fix must include all seven states.

Oct 16

Arizona moves to join

CONGRESS

Money to keep paying for savings

Seventy groups asked Congress for at least $2 billion to extend conservation programs.

$2B

requested

THE NEXT RULES

Talks restart by October 2027

The two-year rules end with 2028. Interior must start the next set by October 1, 2027.

2028

current rules end

FIRST TO WATCH

The April 2027 runoff forecast sets how much water leaves Lake Powell next year.

Sources: Operating Guidelines, sections 1, 5.1, and 5.7; Castle et al. (June 1, 2026); KUNC; the Guardian (October 17, 2026) and (October 1, 2026); Post Independent on the funding letter.

SOURCES AND METHOD

Every chart in this readout is drawn from public data and published research

FEDERAL
Decision and data

Interior, Reclamation, and CRS

Record of Decision; Operating Guidelines; reservoir data; 2025 accounting report; Glen Canyon memo; CRS R45546; WAPA; Arizona Governor.

RESEARCH
Peer reviewed and independent

University researchers and journals

Castle et al., Schmidt et al.; Milly and Dunne; Udall and Overpeck; Castle 2014; Abdelmohsen 2025; JAMA Network Open; NAU; UC Riverside.

GROUPS
Proposals

NGO comments and letters

seven NGOs; Cooperative Conservation; Pacific Institute; Sonoran Institute; Glen Canyon Institute; Grand Canyon Trust.

REPORTING
News and essays

Coverage used for dates and figures

Bloomberg; the Guardian; CalMatters; Circle of Blue; KUNC; azfamily; PBS; framing from Eric Wilburn.

METHOD
Read with care

Calculations

Combined storage, averages, shares, and sums are Planetary PS calculations. Charts, the map, and the dam diagram are drawn from public data. Pathway status and closing questions are a draft reading for discussion.